A Posteriori Error Estimation and Adaptivity

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REFERENCES

Ainsworth–Oden *A Posteriori Error Estimation in Finite Element Analysis*, Wiley, 2000.

Ern–Guermond *Theory and Practice of Finite Elements*, (Chapter 10) Springer, 2004.

General introduction

Recall: for the weak solution $u \in V$ and its discrete approximation $u_h \in V_h$ we had a prototype *a priori* bound (quasi-optimality)

$$\mathcal{E}_h(u, u_h) := \|u - u_h\|_V \leq C \inf_{v_h \in V_h} \|u - v_h\|_V,$$

which, for Poisson and conforming FEM, gives for example

$$|u - u_h|_{H^1(\Omega)} \leq Ch^k |u|_{H^{k+1}(\Omega)}.$$

This gives the theoretical order of convergence, but **no actual estimate of the error** attained once u_h is computed.

An **a posteriori error estimator** is a quantity $\eta_h = \eta_h(u_h)$ **depending on u_h but not on u**, such that

$$\eta_h(u_h) \simeq \mathcal{E}_h(u, u_h).$$

Formal motivation from stability

Consider general elliptic problem: find $u \in V : a(u,v) = f(v) \quad \forall v \in V$.

At least formally, from coercivity,

$$\alpha \|v\|_V^2 \leq a(v,v) \qquad \forall v \in V,$$

one gets

$$\alpha \|v\|_V \leq \sup_{w \in V} \frac{a(v,w)}{\|w\|_V}.$$

Applying this with $v = u - u_h$,

$$\alpha \mathcal{E}_h(u,u_h) = \alpha \|u - u_h\|_V \leq \sup_{w \in V} \frac{a(u - u_h, w)}{\|w\|_V}.$$

Since $a(u,w) = f(w)$,

$$\alpha \mathcal{E}_h(u,u_h) \leq \sup_{w \in V} \frac{f(w) - a(u_h,w)}{\|w\|_V} = \|R(u_h)\|_{V'},$$

where $R(u_h)$ is the **residual**. The residual bounds the error and depends only on u_h !!

However, in this form it is **typically not computable nor localizable**.

A posteriori error estimator desiderata

1. **Reliability:**

$$\mathcal{E}_h(u, u_h) \leq C_2\, \eta_h \qquad \forall u, \forall u_h.$$

2. **Efficiency:**

$$C_1\, \eta_h(u_h) \leq \mathcal{E}_h(u, u_h)$$

   (or more often a local version up to oscillation / patch terms).

3. **Cheap computability.**

4. **Localization:**

$$\eta_h^2 = \sum_{K \in \mathcal{T}_h} \eta_K^2.$$

Effectivity and adaptivity loop

If both reliability and efficiency hold, then estimator and error are of the same asymptotic order.

A standard measure is the **effectivity index**

$$\mathrm{eff}_h := \frac{\eta_h(u_h)}{\mathcal{E}_h(u, u_h)}.$$

- Reliability implies $\mathrm{eff}_h \geq 1/C_2$.
- Efficiency implies bounded effectivity.

Localization leads naturally to adaptive algorithms:

$$\text{SOLVE} \;\to\; \text{ESTIMATE} \;\to\; \text{MARK} \;\to\; \text{ADAPT.}$$

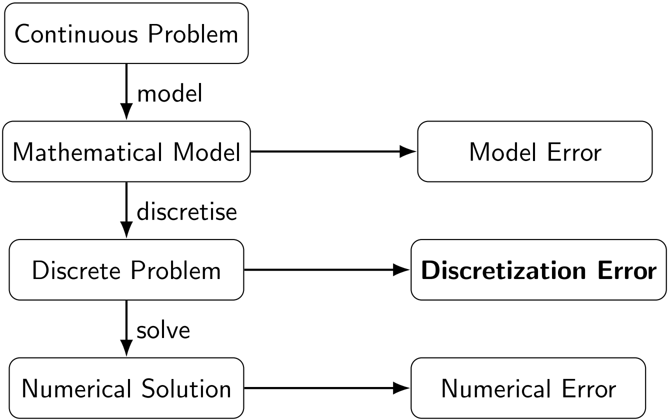
whereby MARK is driven by the local error estimator (indicator) η_K.

What does “adapt” mean?

Adaptivity can mean different things:

- refine / coarsen the mesh (h-adaptivity),
- increase / decrease the polynomial degree (p-adaptivity),
- move nodes / change the mesh geometry (r-adaptivity).

Important: in this lecture we are only assessing the **discretization error**.



Types of estimators

Many different types of a posteriori estimators exist, for example:

- residual-based [Babuška–Rheinboldt, SINUM, 1978],
- hierarchical-type [Bank–Weiser, Math Comp, 1985],
- recovery-type ("smoothing techniques") [Babuška–Rheinboldt, SINUM, 1981],
- goal-oriented / duality-based [Becker–Rannacher, East-West J Num Math, 1996].

Note. Depending on the quantity of interest, different estimators may be preferable.

Recovery-based estimators.

For a conforming FEM approximation of Poisson, the computed gradient is discontinuous across element interfaces. A continuous gradient may be reconstructed by postprocessing:

$$Gu_h \approx \nabla u.$$

Then one may define the estimator

$$\eta_h^2 = \int_\Omega |Gu_h - \nabla u_h|^2 \, dx, \qquad \eta_K^2 = \int_K |Gu_h - \nabla u_h|^2 \, dx.$$

Remarks on recovery estimators

A typical construction is based on averaging.

Attractive because:

- often the gradient is the target rather than in the solution itself.
  In elasticity: stresses and strains rather than displacements.
- $\Rightarrow$ a (smoothed) recovered gradient $G(u_h)$ may already be available in a code as post-processing of the computed discontinuous gradient,
- simple post-processing often delivers higher-order approximations $\Rightarrow$ the difference $\nabla u_h - G(u_h)$ is a very good error estimator in practice. In the case of Poisson

$$\mathcal{E}_h(u, u_h) = \|\nabla u - \nabla u_h\|_{L^2(\Omega)} \approx \|G(u_h) - \nabla u_h\|_{L^2(\Omega)} =: \eta_h^2$$

- they are obviously localizable,

Main weakness: they are often *heuristic*: they may work very well in practice, but do not automatically provide a true guaranteed upper bound for the error (no reliability/efficiency).

1D example and superconvergence (heuristic)

Consider $\Omega = [0, 1]$ and linear FEM. Reconstruct $G(u_h) \in V_h$ by averaging, for instance by using midpoint values.

1D example and superconvergence (heuristic)

Superconvergence result [Zlamal, Math Comp, 1978]:

$$|u_h - I_h u|_{H^1(\Omega)} \leq Ch^2 |u|_{H^3(\Omega)},$$

while the linear FEM error estimate is only

$$|u - u_h|_{H^1(\Omega)} \leq Ch|u|_{H^2(\Omega)}.$$

Moreover, for eg. $v \in \mathbb{P}_2$,

$$G(I_h v) = I_h(v').$$

Hence the interpolation of u is superconvergent toward u_h, suggesting that

$$G(u_h)$$

may[1] superconverge to u'.

[1] Can be made rigorous in this case [Ainsworth–Oden, Wiley, 2000].

2D square-mesh case

On a square mesh in $\mathbb{R}^2$, one may average neighboring gradients to construct a recovered gradient at suitable points.

The Kelly estimator

This leads heuristically to estimators equivalent to jump-based forms such as the Kelly estimator [Kelly, Gago, Zienkiewicz, Babuška, IJNME, 1983]:

$$\eta_K^2 = \frac{h_K}{24} \int_{\partial K} \left[\frac{\partial u_h}{\partial n} \right]^2 ds.$$

Equivalently, one may write it as a sum over faces,

$$\eta_K^2 \sim \sum_{F \subset \partial K} \frac{h_F}{24} \left\| \left[\frac{\partial u_h}{\partial n} \right] \right\|_{L^2(F)}^2 .$$

- The particular coefficient $1/24$ comes from a special-case heuristic choice on square meshes and from enforcing the appropriate gradient-recovery form.
- Kelly's original derivation is different. It is a residual-type estimator. The jump of the normal derivative is the interelement residual obtained by integrating the PDE by parts elementwise (see later).

The ZZ patch recovery estimator

In general, analysis of recovery estimators is based on superconvergence results of the type

$$\|\nabla u - G(I_h^k u)\|_{L^2(K)} \leq C h_K^{k+1} |u|_{H^{k+2}(\omega_K)},$$

where ω_K is a patch around K....

A widely used, technique is the **Zienkiewicz–Zhu superconvergent patch recovery (SPR)** or **ZZ** [Zienkiewicz–Zhu, IJNME, 1992]:

fit, in the least-squares sense or by local averaging, a polynomial on a patch around each vertex using the gradients from neighboring elements.

This gives a recovered gradient field and therefore a practical local error indicator of the form:

$$\eta_{ZZ}^2 = \sum_{K \in \mathcal{T}_h} \|G_h(u_h) - \nabla u_h\|_{L^2(K)}$$

Residual-based error estimator

Residual-based estimators:

- Introduced in [Babuška–Rheinboldt, SINUM, 1978], [Demkowicz–Oden–Strouboulis, Comput Methods Appl Mech Engrg, 1984]
- Analysis (reliability & efficiency) in [Verfürth, J. Comput. Appl. Math., 1994]

Let V Hilbert, $V_h \subset V$, and assume Lax–Milgram.

$$u \in V : \quad A(u,v) = F(v) \quad \forall v \in V$$

$$u_h \in V_h : \quad A(u_h,v_h) = F(v_h) \quad \forall v_h \in V_h$$

Galerkin orthogonality:

$$A(u-u_h, v_h) = 0 \quad \forall v_h \in V_h$$

We'd like to measure how much u_h fails to satisfy the problem in V...

The residual

Recall the definition of the residual $R \in V'$:

$$\langle R, v\rangle = f(v) - a(u_h, v) \quad \forall v \in V$$

Then

$$\|R\|_{V'} = \sup_{v \in V} \frac{\langle R, v\rangle}{\|v\|_V}$$

Key observation:

- R depends only on u_h
- can be used to estimate the error, but in general is not computable...

Step 1: Residual behaves like the error

Going back (using the weak problem):

$$\langle R, v \rangle = a(u - u_h, v)$$

Using continuity of a, that is $a(v, w) \leq \gamma \|v\|_V \, \|w\|_V$

$$\|R\|_{V'} = \sup_{v \in V} \frac{a(u - u_h, v)}{\|v\|_V} \leq \gamma \|u - u_h\|_V$$

Also, using coercivity of a and the definition of the dual norm,

$$\alpha \|u - u_h\|_V^2 \leq a(u - u_h, u - u_h) = \langle R, u - u_h \rangle \leq \|R\|_{V'} \|u - u_h\|_V$$

Therefore residual and error are equivalent:

$$\boxed{\|u - u_h\|_V \simeq \|R\|_{V'}}$$

Step 2: Express residual locally (Poisson)

From now on, restric to the case of Poisson problem, conforming FEM.

Using Galerkin orthogonality:

$$\langle R, v \rangle = a(u - u_h, v) = a(u - u_h, v - v_h) \quad \forall v_h \in V_h$$

Then:

$$\langle R, v \rangle = f(v - v_h) - a(u_h, v - v_h)$$

Element and jump residuals

From

$$\langle R, v \rangle = f(v - v_h) - a(u_h, v - v_h) = \int_\Omega f(v - v_h) - \int_\Omega \nabla u_h \cdot \nabla (v - v_h),$$

integrating by parts element-wise gives

$$\langle R, v \rangle = \sum_K \int_K (f + \Delta u_h)(v - v_h) - \sum_e \int_e \left[\frac{\partial u_h}{\partial n} \right] (v - v_h)$$

Define:

- element residual:
$$f + \Delta u_h$$
- flux jump:
$$\left[\frac{\partial u_h}{\partial n} \right]$$

Bounding the residual

Using Cauchy–Schwarz:

$$|\langle R, v\rangle| \leq \sum_K \Big(\|f + \Delta u_h\|_{0,K} \|v - v_h\|_{0,K} + \|[\partial_n u_h]\|_{0,\partial K} \|v - v_h\|_{0,\partial K} \Big)$$

We need to control $v - v_h$.

Interpolation issue

Problem:

- $v \in H^1$
- pointwise interpolation not well-defined

Solution:

- use interpolation based on averages
- e.g. Clément or Scott–Zhang operator

Clément interpolation

Define:

$$\mathcal{C}_h : H^1(\Omega) \to V_h$$

$$\mathcal{C}_h v = \sum_i (P_i v)\, \phi_i$$

where:

- $P_i v =$ local average over patch
- $\phi_i =$ basis functions

Key estimate:

$$\|v - \mathcal{C}_h v\|_{0,K} \leq C h_K |v|_{H^1(\tilde{K})}$$

where $\tilde{K}$ is the patch of elements neighbouring K.

Also, by trace inequality: $\|w\|_{0,\partial K} \leq C\left(h_K^{-1/2}\|w\|_{0,K} + h_K^{1/2}|w|_{H^1(K)}\right)$

we get $\|v - \mathcal{C}_h v\|_{0,\partial K} \leq C h_K^{1/2} |v|_{H^1(\tilde{K})}$.

Final estimator

Substitute $v_h = \mathcal{C}_h v$:

$$\|R\|_{V'} \leq C \left(\sum_K \left(h_K^2 \|f + \Delta u_h\|_{0,K}^2 + h_K \|[\partial_n u_h]\|_{0,\partial K}^2 \right) \right)^{1/2}$$

Define:

$$\eta_K^2 = h_K^2 \|f + \Delta u_h\|_{0,K}^2 + h_K \|[\partial_n u_h]\|_{0,\partial K}^2$$

Reliability theorem

Theorem (Reliability)

$$\|u - u_h\|_{H^1(\Omega)} \leq C \left(\sum_K \eta_K^2 \right)^{1/2}$$

Key features:

- depends only on computable quantities
- sum of local contributions
- usable for adaptivity

May be pessimistic....

Efficiency (local)

Theorem (Efficiency, [Verfürth, J. Comput. Appl. Math., 1994]**)**

$$\eta_K(u_h) \leq C\big(\|u - u_h\|_{H^1(\omega_K)} + h_K \operatorname{osc}_K\big)$$

Proof:

- bubble functions
- local arguments

Conclusion:

- correct order
- local sharpness
- estimator depends on f, u_h, h
- must use sufficiently accurate quadrature
- extends to many other problems (general elliptic problems, saddle point problems like Stokes,...)